ACT Mathematics

ACT մաթեմատիկա

COURSE CODE	MTH-ACT
PROGRAM	Advanced Placement and Examination Preparation · Առաջադեմ և քննական պատրաստություն
LEVEL	Typically Grades 10–12 · enhanced ACT format
DURATION	48–60 instructional hours (approx. 40 lessons) plus practice testing
PREREQUISITE	Algebra I and Geometry completed; Algebra II completed or in progress.
LEADS TO	Continued practice testing, or SAT Mathematics (MTH-SAT) if the student is comparing formats

COURSE DESCRIPTION

Դասընթացի նկարագրություն

This course prepares a student for the mathematics section of the enhanced ACT. The section is 45 questions in 50 minutes — of which 41 are scored and 4 are unscored field-test items — with four answer choices per question and a calculator permitted throughout. Unlike the SAT, the ACT is linear rather than adaptive: every student sees the same questions in the same order, and the questions trend from easier to harder. The content range is broader than the SAT's and reaches further into Algebra II and trigonometry, but the individual questions are more direct.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

The ACT rewards breadth and speed; the SAT rewards depth and interpretation. Վարժարան prepares students for the ACT accordingly: content coverage is wider and shallower, and pace is trained explicitly, because at roughly sixty-seven seconds per question a student who is accurate but slow will score below their ability. The linear ordering is an advantage that most students fail to use — early questions are reliably easier, so banking time in the first third to spend in the last third is a strategy that can be practised rather than merely described. As with all Վարժարան examination courses, every practice section is followed by item-level error classification, and instruction is sequenced from that analysis.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Computes accurately and quickly without a calculator for routine arithmetic	Reaching for the calculator on simple computation is the most common source of lost time.
Solves linear and quadratic equations fluently	The bulk of the tested algebra; weakness caps the score.
Knows the standard geometry formulas from memory	The ACT provides no formula sheet — a critical difference from the SAT.
Has basic familiarity with right triangle trigonometry	Trigonometry appears on every ACT and is often the last content students acquire.
Has taken at least one full official practice test	Provides the baseline the course is sequenced from.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

1. Perform number and quantity work: integers, fractions, decimals, percentages, exponents, and radicals.
2. Solve linear and quadratic equations, inequalities, systems, and function problems fluently.
3. Work with polynomial, rational, radical, exponential, and logarithmic expressions.
4. Apply plane geometry: angles, triangles, quadrilaterals, polygons, and circles.
5. Apply coordinate geometry: lines, slopes, distance, midpoint, circles, and conic recognition.
6. Compute volume and surface area of standard solids.
7. Apply right triangle trigonometry, the unit circle, trigonometric graphs, and basic identities.
8. Interpret statistics and probability: measures of centre, spread, counting, and probability.
9. Work with matrices, sequences, complex numbers, and vectors at the level the ACT tests.
10. Manage a 45-question, 50-minute section with deliberate pacing and triage.
11. Use a calculator strategically rather than reflexively.
12. Diagnose one's own errors by category and direct practice accordingly.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 4-5 LESSONS

Diagnostic, Format, and Pacing

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TOPICS COVERED

- The enhanced ACT math format: 45 questions in 50 minutes, 41 scored
- Four answer choices and what that changes about guessing and elimination
- The linear format and the easy-to-hard ordering
- No formula sheet is provided: what must be memorized
- Calculator policy and the approved device list
- Full diagnostic test and item-level error analysis
- Classifying errors: content gap, strategic error, timing error, careless error
- The 67-second budget and checkpoint pacing
- Banking time in the first third to spend in the last third
- Triage: attempt, mark, or guess-and-move
- Elimination strategy with four choices
- Building a personal score plan from the diagnostic

LEARNING OBJECTIVES

1. Explain the format and its strategic implications, including the linear ordering.
2. Classify every diagnostic error into one of four categories.
3. Execute checkpoint pacing across a full timed section.
4. Produce a personal score plan naming three priority content areas.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Classifies own errors accurately in 9 of 10 items.
- Reaches each pacing checkpoint within 60 seconds of target on a full section.
- Produces a written personal score plan with named priorities.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Time is spent evenly across the section.	The questions trend harder. Move quickly through the first third to bank time for the last.
Formulas are assumed to be provided as on the SAT.	The ACT provides no reference sheet. Build a memorized formula list in this unit and test it weekly.
Questions are left blank.	There is no penalty for a wrong answer, and with four choices a blind guess is 25%. Answer everything.

TEACHING NOTE

The memorized formula list built here should be tested at the start of every subsequent lesson. The absence of a reference sheet is the most consequential structural difference from the SAT and the one students most often overlook.

UNIT 2 · 10-11 LESSONS

Number, Quantity, and Algebra

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TOPICS COVERED

- Integer properties, factors, multiples, primes, GCF and LCM
- Fractions, decimals, percentages, and conversions
- Ratios, proportions, rates, and unit conversion
- Exponents, radicals, scientific notation, and absolute value
- Complex numbers and their arithmetic
- Linear equations, inequalities, and systems
- Word problems: age, mixture, distance, work, and cost
- Quadratic equations by factoring, formula, and graphing
- The discriminant and vertex
- Polynomial and rational expressions
- Radical and exponential equations
- Logarithms at the level the ACT tests
- Matrices: operations and determinants
- Sequences: arithmetic, geometric, and pattern recognition

LEARNING OBJECTIVES

1. Solve any tested algebraic equation type accurately within the time budget.
2. Translate ACT word problem templates into equations quickly.
3. Work with complex numbers, matrices, and sequences at the tested level.
4. Select between algebraic solution, back-solving, and number substitution.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers this content area at 90% accuracy within the time budget.
- Selects the fastest method in 8 of 10 items.
- Solves ACT word problem templates correctly in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Every problem is solved algebraically.	Back-solving from the answer choices and substituting convenient numbers are often faster. With four choices, back-solving is particularly efficient.

MISCONCEPTION	RECOMMENDED RESPONSE
Complex numbers, matrices, and sequences are neglected because they appear infrequently.	Each appears on essentially every test. A few questions is a meaningful score difference at the top.
Word problems are read once and attempted immediately.	Identify the asked quantity and the given relationships before writing anything. This costs seconds and saves minutes.

TEACHING NOTE

The ACT tests a wider content range than the SAT but at shallower depth. A student with broad coverage and moderate depth outperforms one with deep algebra and gaps elsewhere — sequence practice with that in mind.

UNIT 3 · 7-8 LESSONS

Functions and Advanced Topics

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TOPICS COVERED

- Function notation, evaluation, and composition
- Domain and range
- Inverse functions
- Graphs of linear, quadratic, polynomial, exponential, logarithmic, and rational functions
- Transformations of graphs
- Piecewise functions
- Asymptotes and end behaviour
- Modelling with functions
- Interpreting a graph in context
- Systems involving nonlinear functions
- Vectors: components, magnitude, and operations
- Logic and reasoning items

LEARNING OBJECTIVES

1. Evaluate, compose, and invert functions fluently.
2. Identify a function type from its graph and vice versa.
3. Apply transformations to a parent function and predict the effect.
4. Work with vectors at the tested level.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers function items at 88% accuracy within the time budget.
- Matches graphs to function types in 9 of 10 trials.
- Applies transformations correctly in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
f(x) is read as multiplication under time pressure even when the notation is understood.	Practice at speed. Notation errors that do not occur in untimed work reappear under pressure and must be trained out.
Composition order is reversed.	f(g(x)) applies g first. Write the inner evaluation explicitly rather than composing mentally.
Transformations of horizontal direction are applied intuitively.	f(x − 3) shifts right. Verify with a point rather than reasoning from the sign.

TEACHING NOTE

Graph-matching questions are among the fastest on the test for a prepared student and among the slowest for an unprepared one. Drill recognition of the standard graph shapes until identification is instantaneous.

UNIT 4 · 9–10 LESSONS

Geometry

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TOPICS COVERED

- Angles: complementary, supplementary, vertical, parallel lines and transversals
- Triangles: angle sum, exterior angles, similarity, congruence, inequality
- The Pythagorean theorem and special right triangles
- Quadrilaterals and polygons: properties, angle sums, area
- Circles: circumference, area, arcs, sectors, chords, tangents, inscribed angles
- Area and perimeter of composite figures
- Volume and surface area of prisms, cylinders, cones, pyramids, and spheres
- Coordinate geometry: distance, midpoint, slope, parallel and perpendicular lines
- Equations of lines and circles
- Transformations in the coordinate plane
- Recognizing conic sections
- Three-dimensional reasoning and cross-sections

LEARNING OBJECTIVES

1. Apply all standard geometry formulas from memory without a reference sheet.
2. Solve multi-step geometry problems combining two or more relationships.
3. Work fluently in coordinate geometry including circles.
4. Reason about three-dimensional figures and their cross-sections.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers geometry items at 90% accuracy within the time budget.
- Recalls all required formulas correctly on a weekly formula check.
- Solves multi-step geometry problems in 4 of 5 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Formulas are half-remembered — for instance the cone volume without the one-third factor.	The weekly formula check exists for this. Partial recall is worse than no recall because it produces confident errors.
Diagrams are assumed to be to scale.	The ACT usually draws to scale but does not guarantee it. Use the diagram to estimate and eliminate, and the given information to compute.
Composite figures are attacked without decomposition.	Decompose into standard shapes and mark each piece before computing anything.

TEACHING NOTE

Geometry carries more weight on the ACT than on the SAT, and the absence of a reference sheet makes formula recall a genuine scoring factor. The weekly formula check is not optional in this course.

UNIT 5 · 7-8 LESSONS

Trigonometry, Statistics, and Probability

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TOPICS COVERED

- Right triangle trigonometry: the six ratios
- Solving right triangles and applied trigonometry problems
- The unit circle and radian measure
- Graphs of sine, cosine, and tangent: amplitude and period
- Basic trigonometric identities
- The Law of Sines and the Law of Cosines
- Trigonometric equations at the tested level
- Mean, median, mode, range, and weighted averages
- Standard deviation conceptually and comparatively
- Reading tables, charts, and graphs
- Counting principles, permutations, and combinations
- Probability of single and compound events
- Expected value

LEARNING OBJECTIVES

1. Solve right triangle trigonometry problems quickly and accurately.
2. Identify amplitude and period from a trigonometric graph or equation.
3. Compute weighted averages and reason about the effect of a data change.
4. Apply counting principles and compute probabilities of compound events.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Answers trigonometry items at 88% accuracy within the time budget.
- Computes weighted averages correctly in 9 of 10 trials.
- Applies counting and probability correctly in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Trigonometry is skipped in preparation because it appears late in the test.	It appears on every ACT and typically at a scoring-relevant density. Neglecting it caps the achievable score.
Weighted averages are computed as simple averages.	Weight by the count in each group. This is a recurring and very learnable ACT item type.
Permutations and combinations are chosen without considering order.	Ask whether rearrangement produces a different outcome. If yes, order matters.

TEACHING NOTE

Trigonometry is where a student targeting a score above 30 gains the most, and it is the area most often left until too late. Sequence it earlier than instinct suggests for high-target students.

UNIT 6 · 5-6 LESSONS PLUS SCHEDULED FULL-LENGTH TESTS

Timed Practice, Strategy, and Review

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TOPICS COVERED

- Full-length official practice tests under authentic timed conditions
- Checkpoint pacing across all 45 questions
- Banking time early and deploying it late
- Triage decisions within fifteen seconds
- Elimination with four answer choices
- Strategic calculator use and when not to reach for it
- Item-level error analysis and error log maintenance
- Targeted remediation assigned from the error log
- Weekly formula recall checks
- Managing fatigue across the full test
- Test-day logistics and permitted materials
- Score interpretation and superscoring considerations

LEARNING OBJECTIVES

1. Complete a full 45-question section within 50 minutes with no blanks.
2. Triage within fifteen seconds and act on the decision.
3. Maintain an error log and derive the next week's practice from it.
4. Execute a stable, rehearsed test-day routine.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Completes full sections within time with no blanks in 3 consecutive practice tests.
- Reduces the dominant error category by half from the diagnostic baseline.
- Demonstrates a stable or rising score across the final three practice tests.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Too long is spent on a hard early question.	All questions score equally. Mark and move; the linear ordering means later questions are harder, not more valuable.
The calculator is used for computation that is faster by hand.	Reaching for the device costs seconds each time. Practice recognizing when mental arithmetic is faster.
Practice tests are taken without item-level analysis.	The test measures; the analysis teaches. Without the analysis the practice test is largely wasted.

TEACHING NOTE

Space full practice tests every two to three weeks. Improvement comes from the remediation between tests; a student taking a test every week has no time to fix anything and typically plateaus.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Diagnostic Practice Test	Before instruction begins	A full official practice test under authentic conditions; produces the item-level baseline that sequences the course.
Unit Check	End of each content unit · 25 min	Timed content-specific sets in ACT format and pacing.
Formula Recall Check	Every lesson from Unit 1 · 3 min	The memorized formula list; the ACT provides no reference sheet, so this is a genuine scoring factor.
Error Log Review	Every lesson	Ongoing classification of errors into content, strategic, timing, and careless categories.
Full Practice Test	Every 2–3 weeks	Official practice tests under authentic timed conditions, followed by item-level analysis.
Examination Format	Enhanced ACT · Mathematics section	45 questions in 50 minutes — approximately 67 seconds per question. 41 questions are scored; 4 are unscored field-test items distributed through the section. Four answer choices per question. The section is linear: all students see the same questions in the same order, trending from easier to harder. A calculator is permitted for the entire section, including an approved graphing calculator or the built-in Desmos calculator on the digital form. No formula reference sheet is provided.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 60–75 min	12–16 weeks including practice tests — the most common track.
Intensive	2 lessons/week	8 weeks; suitable when the test date is close and the baseline is already reasonable.
Extended Mastery	1 lesson/week over two testing cycles	24–30 weeks, combining ACT preparation with genuine content remediation.
Summer Program	3 lessons/week, 6 weeks	Full course before an autumn test date.
Retest Track	1 lesson/week, 6 weeks	Targeted work derived entirely from the previous official score report.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Forty-five to sixty minutes per lesson: one timed ACT-format set matching the current content area, plus remediation problems assigned from the error log, plus formula list review. Every timed set is self-analyzed before the next lesson, with each missed item classified by error type and the correct method written out. Timing is recorded per question so that speed can be diagnosed separately from accuracy — on a test with sixty-seven seconds per question, these are genuinely different problems requiring genuinely different remedies.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- Established a diagnostic baseline with item-level error classification.
- Memorized the complete formula list and passes weekly recall checks.
- Executes checkpoint pacing across a full timed section.
- Solves all tested number, quantity, and algebra content within budget.
- Works with complex numbers, matrices, and sequences.
- Handles functions, graphs, transformations, and vectors.
- Applies all plane, solid, and coordinate geometry from memory.
- Solves right triangle trigonometry and works with trigonometric graphs.
- Applies statistics, counting, and probability.
- Completes 45 questions in 50 minutes with no blanks.
- Reduced the dominant error category by half from baseline.
- Achieved the target score on a full official practice test.
- Completed ACT Mathematics preparation — test ready.

MATERIALS AND RESOURCES

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- Official ACT practice tests — the primary practice material.
- Approved graphing calculator, or the digital form's built-in calculator.
- Վարժարան ACT formula list — memorized, tested weekly, since no reference sheet is provided.
- Վարժարան ACT error log template with per-question timing.
- Վարժարան ACT problem bank — approximately 1,800 items tagged by content area, difficulty, and target solution time.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.